

Rickey Patel

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D.Eng. Candidate · AI & Workforce Analytics · Penn State University

PROFILE

Doctoral candidate researching how artificial intelligence is reshaping the workforce. I build production-grade cloud pipelines that turn large-scale employer demand data into longitudinal evidence of skill shifts, mapped to O*NET competency frameworks, and apply NLP, spatial, and statistical methods to identify who risks being left behind in the digital economy. A background in specialty-pharmacy patient services and research in clinical AI stewardship (fairness auditing, explainable AI) shape an equity-centered approach to every system I design. NABET 2025 Best Paper Award recipient.

EDUCATION

- Doctor of Engineering (D.Eng.) | The Pennsylvania State University** Expected 2027
Research focus: AI & Workforce Analytics · Advisor: Dr. Satish M. Srinivasan
- Graduate Certificate in Engineering Leadership & Innovation Management (in progress).
- Master of Science, Information Science | The Pennsylvania State University** Aug 2025
GPA 3.73
- Capstone: real-time clinical decision support using deep learning, NLP, Kafka, Flink, Spark, and HDFS.
- Bachelor of Science, Information Sciences & Technology | Penn State Abington** May 2021
GPA 3.89 · Concentration: Human Physiology & Biology

RESEARCH & TEACHING

- Research Assistant to Dr. Satish M. Srinivasan | Penn State University** Current
- Lead the design and operation of GCP-native data pipelines tracing AI-driven skill demand across CS, MBA, Data Analytics, AI, healthcare, and faith-based labor markets.
 - Co-authored the NABET 2025 Best Paper and two papers accepted to NABET 2026; produced network diagrams and comparative visualizations for Yang & Srinivasan (2026) on marketing skill networks.
 - Prepare grant and funding materials (demo scripts, briefing decks, quick-reference guides) positioning the pipeline as a running research program with peer-reviewed output.
- Course Developer — CSC 894: Healthcare AI Capstone | Penn State Great Valley** 2026
- Developing a graduate capstone built around HealthRiskAI, a healthcare application that processes claims data through a GCP pipeline with HIPAA compliance emphasized throughout.
 - Revised all ten lessons with video scripts, labs, pseudocode, and exercises; built a Clinical Stewardship & Design Equity Rubric requiring fairness audits and XAI ethics-by-design at every milestone.

FEATURED ENGINEERING

- Job Postings Intelligence Pipeline | GCP · In Production** 2025 – Present
USAJobs + Jooble → Pub/Sub → Dataflow → BigQuery → Vertex AI AutoML
- Collects, scores, and maps postings across CS, MBA, Data Analytics, and AI domains to O*NET 24.2 competency frameworks; registered a domain classifier model in Vertex AI.
 - Migrated from a Colab prototype to fully GCP-native services (Cloud Run, Cloud Scheduler, Secret Manager, GCS) with three deduplication layers and partitioned BigQuery tables, at under \$600/year in infrastructure cost.
 - Built a ROAR HPC variant for Penn State's supercomputer — DuckDB, a Streamlit dashboard, and SLURM scheduling — achieving full parity with zero cloud dependencies.
- Religious Job Listings Dataset (RJLD) | Independent Research** 2024 – Present
Multi-source IRS, Census, USDA, SerpAPI, USAJobs, and Jooble pipeline for spiritual-facilitation roles
- Fine-tuned a RoBERTa multi-label classifier gated on inter-annotator agreement ($\kappa \geq 0.6$); applied BERTopic, LDA ($k = 5$, $c_v = 0.3851$), OLS regression, and Moran's I spatial autocorrelation.
 - Identified a digital equity gap: digital labor ratio correlates inversely with median household income ($r = -0.402$, $p = 0.038$) across 27 states. SSRC grant application and peer-reviewed publication in progress.

- Designing a stream-first reference architecture combining continuous fairness auditing (demographic parity, equalized odds), XAI modules (SHAP, natural-language rationales), and HIPAA/GDPR-compliant model lifecycle management; audit protocols target sepsis, readmission, and diagnostic classification tasks.

PRESENTATIONS & PUBLICATIONS

- **Lightning Talk, NJBDA 2026 Annual Symposium** — “Architectural Patterns and Orchestration Frameworks in Multi-Agent AI for Healthcare.”
- **NABET 2026 (accepted)** — “Building Trust in AI: Ethical Deployment and Regulatory Oversight in Healthcare.”
- **NABET 2026 (accepted)** — “Impact and Current State of Artificial Intelligence in the Religious Market.”
- **NABET 2025, Best Paper Award** — “Impact of Artificial Intelligence in the Healthcare Sector,” with S. M. Srinivasan.

TECHNICAL SKILLS

Cloud & Data Engineering: GCP (Pub/Sub, Dataflow/Apache Beam, BigQuery, Cloud Run, Cloud Scheduler, GCS, Secret Manager, Vertex AI), Kafka, Flink, Spark, HDFS, DuckDB, SLURM (Penn State ROAR), GitHub Actions

Languages: Python, R, SQL, JavaScript (Node.js, React Native)

NLP & Machine Learning: BERTopic, LDA, RoBERTa fine-tuning, deep learning, scikit-learn, statsmodels, XAI (SHAP, saliency maps)

Statistical & Spatial: OLS regression, Pearson correlation, Moran's I (esda, libpysal), GeoPandas, network analysis (betweenness centrality, structural holes)

Labor Market Data: O*NET 24.2, USAJobs API, Jooble API, SerpAPI Google Jobs, IRS, Census, USDA

Visualization: Streamlit, Plotly, Chart.js, matplotlib/seaborn, Power BI

Healthcare & Governance: HIPAA, HL7 FHIR, FDA SaMD guidance, fairness auditing, data governance

PROFESSIONAL EXPERIENCE

Patient Services Associate — Patient Care Coordinator | Asembia / ASPN Pharmacies,
Trevose, PA

Jul 2020 – Apr 2024

- Coordinated specialty-medication access among patients, providers, insurers, and pharmacy services in full HIPAA compliance, managing a high-volume caseload.
- Reduced prescription processing time by 30% and prior-authorization turnaround by 25%; increased patient retention by 15%.
- Maintained 100% data-entry accuracy across internal compliance audits; mentored and onboarded new hires.

Customer Service Supervisor | Wawa, Inc., North Wales & Roslyn, PA

Sep 2018 – Jul 2021

- Cut register variances 25% with a dual-verification process; contributed to a 12% lift in targeted seasonal sales; trained 10+ associates, shortening onboarding by about one week per hire.

HONORS, CERTIFICATIONS & SERVICE

- NABET 2025 Best Paper Award.
- Certificates, Penn State Great Valley (June 2025): Business Process Integration; Cyber Threat Analytics and Prevention.
- Data Science Volunteer, Montgomery County Public Safety Department (Jul 2020) — built Power BI dashboards supporting COVID-19 contact tracing.
- Earlier academic projects: association-rule mining for curriculum optimization (47 → 34 electives in 4 tracks); Community Rides React Native app; RSA cryptosystem and cyber forensics.